

OMKAR GOCHADE

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Leetcode: <https://leetcode.com/u/Omiipk/> | AtCoder: <https://atcoder.jp/users/Omiipk>

Innovative Technologist specializing in software engineering, machine learning, data science, and data structures and algorithms with a passion for finding solutions to untapped blocks and real-world problems at the intersection of AI, quantitative systems, and scalable infrastructure. My expertise lies in bridging advanced mathematics with cutting-edge engineering (Python, cloud systems, big data computing to build robust AI/ML solutions. My research at the International Conference of Advancements in Mathematics and Data Science (NIT Bhopal) and hands-on projects in cybersecurity, cloud computing, and generative AI reflect my commitment to pushing boundaries. With a foundation in software engineering, econometrics, and data science, I bring a unique perspective to machine learning engineering, ensuring models are not just theoretically sound but deployable, efficient, and actionable.

SKILLS:

Languages – Python, C++, SQL, SQLite3, R, MATLAB, HTML.

Tools, Platforms, Frameworks- Google Collab, PyCharm, Roboflow, DB Browser, DataStax Astra DB, Django, Langflow, Label Studio, Tableau, NVIDIA NGC Catalogue, IBM Watson, TensorFlow, OpenCV, PyTorch, YOLO, Keras, Scikit-Learn.

Core Competencies and Technical Skills – Data Science and Analysis, Data Structures and Algorithms, Operating Systems, Computer Networks, Probability and Statistics, Linear Algebra, DBMS, Computer Architecture, Neural Networks, Generative AI, Prompt Engineering, Feature Engineering, Big Data and Cloud Computing and Data Centre Management, Optimization techniques, Data Mining, Econometrics, Supply and value chain management, Data protection and replication, Distributed systems.

Soft Skills- Professional Communication (English, Hindi, Marathi), Public Relations and Customer Satisfaction, Multitasking and Decision Making, Teamwork and Collaboration, Product and Media Management.

EDUCATION:

Maulana Azad National Institute of Technology, Bhopal, India: B.Tech in Mathematics and Data Science

Course Duration: 2022- 2026. CGPA: 8.40 (Latest SGPA: 9.14)

Shri Shivaji Vidya Mandir and Junior College, Pune: Higher Secondary Certificate-HSC (Course Duration: 2020-2022)

Percentage: 77.5% (Subjects: Physics, Chemistry, Mathematics, Computer Science)

Mar Ivanios Convent High School, Pune: Secondary School Certificate-SSC (2020)

Percentage: 96.2% (Subjects: Science, Mathematics, Computers)

INTERNSHIPS and PROFESSIONAL WORK EXPERIENCE:

1) IBM – Generative AI and Machine Learning Industrial Training (August 2024 to October 2024, virtual)

- Developed an ML model to predict diseases and severity levels from symptoms, integrated with IBM Watson for intelligent interaction and orchestration. Integrated the solution with IBM Watson's cognitive services to enable intelligent interaction, automated reasoning, and real-time decision support within a cloud environment
- Tech stack: IBM Cloud, Watson Assistant, Python, Scikit-learn, NLP, Watson Studio, Computer Architecture, Neural Networks.

2) FEYNN LABS – Data Science Intern (May 2024 to July 2024):

- Designed and prototyped an ML-driven dynamic pricing system using deep learning (LSTMs, transformers) and survival analysis (Gompertz/Weibull models) to forecast demand, customer lifetime value, and subscription elasticity for SaaS/Aggregator business models
- Developed a scalable Python framework with automated feature engineering and model training pipelines, validated through back testing against historical data.

3) BRAINLOX – Market Research and Analyst (May 2023 to August 2023):

- Drove revenue growth for 20+ financial products by leveraging data analytics, AI tools, and market research to optimize pricing strategies. Conducted behavioral analysis via campaigns and surveys to refine demand forecasting and price elasticity modeling.
- Diagnosed and resolved a 2-month sales stagnation by realigning pricing policies and streamlining processes, resulting in 10x customer growth, 3x transaction volume, and reduced revenue leakage.

Projects:

1) Multi-Device Broker for Cloud Computing: Designed and implemented a Python-based cloud broker system enabling cross-platform access (mobile/laptop/IoT) to restricted cloud services through runtime protocol transformation. Developed device registration with credential management, dynamic protocol translation gateways (JSON/XML/MQTT), and metadata-driven mapping logic to resolve hardware-service compatibility. Tech stack and Skills: Python, Cloud middleware solution, Systems Design, Device lifecycle management, service, and data management. [Link](#)

2) AI AGENT for retail fraud detection and transactions: Designed an AI Agent, an all-in-one digital shield for secure transactions, identity protection, real-time threat detection, and an AI-powered fraud Detection and Prevention System for the Digital Payment Channel. Frauds handled by this agent are Fake Payment Links, Vishing (Voice Phishing), Request Money Scam, Fake Customer Care Numbers, Remote Access Scams, QR Code Scam, and Loan & Cashback Fraud. Built solutions like Analysis of Smart Messages by LLMs, Distance Metric Based Fake Link Detection, Validate Website using external APIs, Access Control, Notification to User, etc. Tech Stack and skills: Python, machine learning (Random Forest, Xgboost, SVM, NLP), Data augmentation and preprocessing, Cybersecurity. [Link](#)

3) KAWACH (Knowledge-based Advanced Watchdog for Authentication, Cybersecurity, and Heuristics): As part of an advanced cybersecurity initiative, I designed and prototyped a multi-layered file protection system leveraging deceptive defence mechanisms. The system strategically deploys "F-B" landmine files around sensitive directories and files "F-A", which trigger upon unauthorized access to generate multiple "FC" decoy files—near-perfect replicas with 60-90% content alteration via NLP augmentation (NLTK/SpaCy) and GANs. Each F-C file is accompanied by a password-locked metadata log recording all modifications, ensuring full recoverability even if the original files are compromised. To further trap attackers, I implemented a unique "Lock-in Lock-out" authentication protocol requiring dynamic passwords for both file access and navigation between files.

Tech Stack and skills: Python, NLP, Generative Adversarial Networks (GANs), cybersecurity, data structures, file management, security pipelines. [Link](#)

4) Automated Research and Trigger Finder (ART Finder): Developed ART Finder, an AI-powered tool that automates ad research by scraping data from YouTube and Reddit to identify user pain points and competitor strategies. Leveraged NLP and sentiment analysis to extract insights, high-performing hooks, and CTAs. Built an interactive visualization and validation dashboard using Python, BeautifulSoup, Selenium, and data processing. [Link](#)

5) Feature Extraction from Remote Sensing High-Resolution Data using AI/ML (Ex: High Tension tower, windmill, electric substation, Brick Kiln, farm bunds) Developed an AI/ML-based solution with 42% accuracy for extracting features from high-resolution remote sensing data of over 80 images, targeting objects such as high-tension towers, windmills, and farm bunds. The workflow included data collection, preprocessing, model training, and postprocessing using YOLO for real-time detection and GDAL for precise geospatial data handling. Tech Stack: Python, OpenCV, GDAL, Google Colab, Roboflow, QGIS. Utilized deep learning frameworks and geospatial data processing libraries. [Link](#)

6) Machine Learning-based Distance Prediction and Fuel & Automobility Management: Developed a model that predicts vehicle travel distance based on fuel levels using Linear Regression, Random Forest, and Gradient Boosting, achieving 82% overall accuracy. This project offers insights into driving habits and user analytics and was published in the International Conference of Recent Advancements in Mathematics and Data Science as a research paper. [Link](#)

7) Data Analysis of Cafe Coffee Day (CCD) @Bhopal (near Lake View): Conducted a comprehensive analysis of over 500 customer reviews and transaction data using Python, implementing statistical models to identify key factors influencing customer behavior and purchasing patterns. Performed pricing analysis and evaluated financial trends across multiple dimensions, including seasonality, demographics, and product categories, to optimize revenue strategies. Developed predictive models to examine customer preferences by age group and seasonal trends, identifying optimal pricing strategies for various market segments. Utilized advanced data visualization techniques with Pandas, NumPy, and Seaborn to create detailed dashboards analyzing parameters such as service quality, ambiance ratings, and taste preferences, resulting in actionable insights that informed strategic pricing decisions and enhanced customer satisfaction metrics. [Link](#)

ACHIEVEMENTS AND CERTIFICATIONS:

- 1) Research Paper on Machine Learning Distance Prediction and Fuel & Automobility Management: Selection at ICRAMDS, MANIT, Bhopal, Madhya Pradesh.
 - 2) Participated in multiple hackathons - ISRO Bhartiya Antariksh Hackathon, Walmart Sparkathon, TATA InnoVent Hackathon, Hack'Ndore, NASA Space Apps Hackathon, Level Supermind Hackathon-team under top 10 from 400 teams.
 - 3) NPTEL Elite+Silver certification in Introduction to Econometrics, Deep learning, Big Data Computing, Accelerated Applied Artificial Intelligence
 - 4) INMO-Indian National Mathematics Olympiad -top 1%: June 2022. Issued by - Greater Bombay Teachers Association, IAPT (Indian Association of Physics Teachers).
 - 5) Find a Leader in You (FLY Program) Certification by Competitive Mindset Institute, USA.
 - 6) Grand Master Certification by SIP Abacus, Pune.
 - 7) IBM Industrial Training Certification in Generative AI and Machine learning.
 - 8) Solved 500+ DSA problems,100+ SQL problems, and 100+ machine learning and data science problems across multiple platforms.
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HOBBIES AND EXTRA-CURRICULAR ACTIVITIES

- 1) Swimming
- 2) Reading Marathi Literature.
- 3) Basketball.
- 4) Travelling.